A combinatorial approach for the synthesis and analysis of $Al_xCr_yMo_zNbTiZr$ high-entropy alloys: Oxidation behavior

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To overcome the limited feasibility of various refractory high-entropy alloys (HEAs) due to the presence of (i) very dense elements (W and Ta), (ii) costly elements (Hf and Ta), and (iii) oxidation prone elements (V) in them, $Al_xCr_yMo_zNbTiZr$ HEAs were prepared via arc-melting. Considering the critical nature of oxidation resistance in high-temperature applications, HEAs were characterized to form a combinatorial library of microstructural and oxidation behavior. $Al_xCr_yMo_zNbTiZr$ HEAs revealed multiphase microstructures consisting of intermetallic phases along with BCC matrices. Mass loss and porous microstructures were obtained in Mo-rich HEAs after oxidation at 1000 °C for 1 h. The presence of Al enhanced the oxidation resistance and developed a protective oxide layer on the HEAs. $Al_{30}Cr_{10}$-NTZ exhibited promising potential for use in high temperature applications, as it showed an oxidation time exponent of ~0.5 and a dense and continuous oxide layer.

I. INTRODUCTION

Refractory alloys with good high-temperature mechanical properties are considered for applications that require substantially higher service temperatures.[1] However, applications of such alloys become limited due to their low ductility at room temperature and poor resistance to oxidation at elevated temperatures and high densities.[1] To overcome these constraints, the metallurgical community is exploring multiprincipal alloys with high mixing entropies, or "high-entropy alloys (HEAs)",[1,2] which are alloys composed of 5–13 elements in equiatomic or near-equiatomic concentrations (5–35 at.%).[3–5] Lattice distortions increase the hardness and strength of these alloys, which form solid solutions over intermetallic phases due to the high mixing entropy.[3,5,6]

The presence of multiple principal elements over a wide range of concentrations has opened a new dimension for researchers who explore new alloys.[4] The development and microstructural/mechanical analyses of new HEAs with enhanced properties are critical for progress towards their potential use in high-temperature applications.[7] Several new single-phase (e.g., HfNbTaTiZr[5]) and multiphase (e.g., CrNbTiVZr[2]) refractory HEAs have been reported, exhibiting promising strength, ductility, toughness, thermal stability, and wear resistance levels at both room temperature and elevated temperatures.[3,7,8] The high-temperature yield strengths observed for some HEAs, which are even better than those of Ni-based superalloys,[2,7,9,10] making them potential candidates in various automotive, aerospace, and nuclear applications. For example, high-temperature (1600 °C) strength capabilities of 405 MPa and 477 MPa were observed for MoNbTaW and MoNbTaVW, respectively.[11] Furthermore, good room-temperature ductility in HfNbTaTiZr has been reported.[12] In the same study, HfNbTaTiZr was modified to form HfMoNbTiZr to reduce the cost and density and improve the strength, which demonstrates the versatility of the HEA design space.[12] The replacement of heavy elements Ta and Hf with lighter elements (i.e., Cr and Mo, respectively) to form $CrMo_{0.5}NbTa_{0.5}TiZr$ can result in a density of 8.2 g/cm^3. The $CrMo_{0.5}NbTa_{0.5}TiZr$ HEA showed good oxidation resistance (higher than Nb-based alloys, AlNbSiTi and MoNbSi) and yield strengths of 2000 MPa and 745 MPa at 25 °C and 1000 °C, respectively.[12]

Because exchanging elements and varying elemental concentrations have significant effects on the properties of refractory HEAs, there is a need for a combinatorial

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library including a wide range of elements and compositions of HEAs so that the most appropriate composition with enhanced properties can be selected for further research and commercialization. In this work, we consider alloys for high-temperature applications. Accordingly, the constituents were selected from refractory metals Cr, Nb, Ti, and Zr owing to their low densities and high melting points.[3] Due to their potential to reduce the density[8] and improve the oxidation resistance of alloys,[2] Al was also selected.[8] Hf, Ta, and W were not selected due to their high densities.[3] Furthermore, vanadium was not selected considering its adverse effects on the oxidation resistance of $Al_{0.5}CrMoNbTiV$.[13] We employ Mo due to its relatively low density compared to Ta and W and its good potential for high strength.[12]

Previous reports suggest the successful utilization of Nb, Ti, and Zr in HEAs at relatively equal concentrations due to the good relative solubility, a result of their propensity to form solid solutions without any intermediate compounds.[14] A number of HEAs containing equiatomic Nb, Ti, and Zr have been reported, including CrNbTiVZr,[2] HfNbTaTiZr, HfMoNbTiZr, $CrMo_{0.5}NbTa_{0.5}TiZr$[12]; this finding motivated us to use equal ratios of Nb, Ti, and Zr.

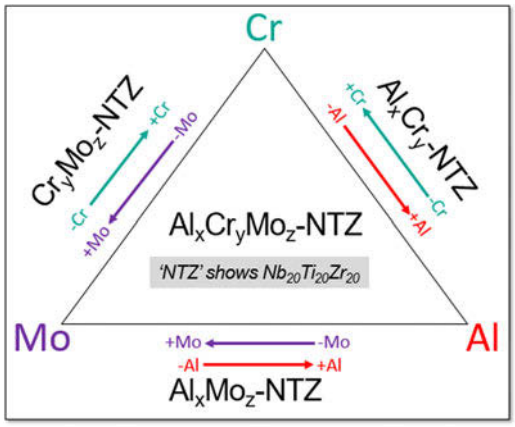


FIG. 1. A combinatorial approach to the development and analysis of the $Al_xCr_yMo_zNbTiZr$ HEA.

It has been reported that varying the Al, Cr, or Mo concentration can have significant effects on the properties of HEAs. $Mo_xNbTiVZr$ (x = 0.3–1.2), $Mo_xNbTiV_{0.3}Zr$ (x = 0.1–1.5),[15] and $HfMo_xNbTaTiZr$ (x = 0–1)[16] all showed improvements in strength, from 1289 to 1765 MPa, 932 to 1576 MPa, and 1015 to 1512 MPa, respectively, due to increases in the Mo content.

As observed in $AlCr_xNbTiV$ (x = 0 to 1.5), increasing the Cr content improved the HEA strength and oxidation resistance.[9] However, the relatively high shear modulus of Cr compared to other transition metals and its high notch sensitivity[17] led to decreased ductility in Cr-bearing HEAs.[18] For example, the fracture strain of AlNbTiV was reduced from 5 to 1% when Cr was added to form $AlCr_{0.5}NbTiV$.[18]

While Al reduces the density and improves the strength[9] and oxidation resistance of refractory HEAs,[2] recent research on the effects of Al in HEAs has shown various results. Adding Al to $CrMo_{0.5}NbTa_{0.5}TiZr$ stabilizes the solid-solution phases and reduces ordered intermetallics.[2] Similar HEAs, such as NbTaTiV, MoNbTiV, and HfNbTaTiZr, also retain their structure after Al inclusion.[2] However, it has been shown that including Al promotes the formation of intermetallics in NbTiVZr.[2]

In this manuscript, we present the development of a combinatorial library of the $Al_xCr_yMo_zNbTiZr$ HEA system, which is henceforth abbreviated as "$Al_xCr_yMo_z$-NTZ", where 'NTZ' refers to equiatomic Nb, Ti, and Zr. The combinatorial approach is shown in Fig. 1.

Oxidation resistance is one of the most critical performance metrics of refractory metals and alloys for high-temperature applications. Therefore, the oxidation behavior should be analyzed to understand the effects of alloying elements on the oxidation resistance capabilities of refractory HEAs. Although there are many reports on the mechanical properties of refractory HEAs,[4,7,19] only a few analyses of refractory HEAs pertain to the oxidation behavior of certain selected HEA systems.[20–24] We discuss the high-temperature oxidation

TABLE I. Detailed compositions of the HEA $Al_xCr_yMo_z$-NTZ.

Sample name			HEA Compositions
$Al_xCr_yMo_z$-NTZ ($Al_xCr_yMo_z$-NTZ (NTZ = $Nb_{20}Ti_{20}Zr_{20}$)		$Al_{20}Cr_{10}Mo_{10}$-NTZ	$Al_{20}Cr_{10}Mo_{10}Nb_{20}Ti_{20}Zr_{20}$
		$Al_{10}Cr_{20}Mo_{10}$-NTZ	$Al_{10}Cr_{20}Mo_{10}Nb_{20}Ti_{20}Zr_{20}$
		$Al_{10}Cr_{10}Mo_{20}$-NTZ	$Al_{10}Cr_{10}Mo_{20}Nb_{20}Ti_{20}Zr_{20}$
	Al_xMo_z-NTZ (y = 0)	$Al_{30}Mo_{10}$-NTZ	$Al_{30}Mo_{10}Nb_{20}Ti_{20}Zr_{20}$
		$Al_{20}Mo_{20}$-NTZ	$Al_{20}Mo_{20}Nb_{20}Ti_{20}Zr_{20}$
		$Al_{10}Mo_{30}$-NTZ	$Al_{10}Mo_{30}Nb_{20}Ti_{20}Zr_{20}$
	Al_xCr_y-NTZ (z = 0)	$Al_{30}Cr_{10}$-NTZ	$Al_{30}Cr_{10}Nb_{20}Ti_{20}Zr_{20}$
		$Al_{20}Cr_{20}$-NTZ	$Al_{20}Cr_{20}Nb_{20}Ti_{20}Zr_{20}$
		$Al_{10}Cr_{30}$-NTZ	$Al_{10}Cr_{30}Nb_{20}Ti_{20}Zr_{20}$
	Cr_yMo_z-NTZ (x = 0)	$Cr_{30}Mo_{10}$-NTZ	$Cr_{30}Mo_{10}Nb_{20}Ti_{20}Zr_{20}$
		$Cr_{20}Mo_{20}$-NTZ	$Cr_{20}Mo_{20}Nb_{20}Ti_{20}Zr_{20}$
		$Cr_{10}Mo_{30}$-NTZ	$Cr_{10}Mo_{30}Nb_{20}Ti_{20}Zr_{20}$

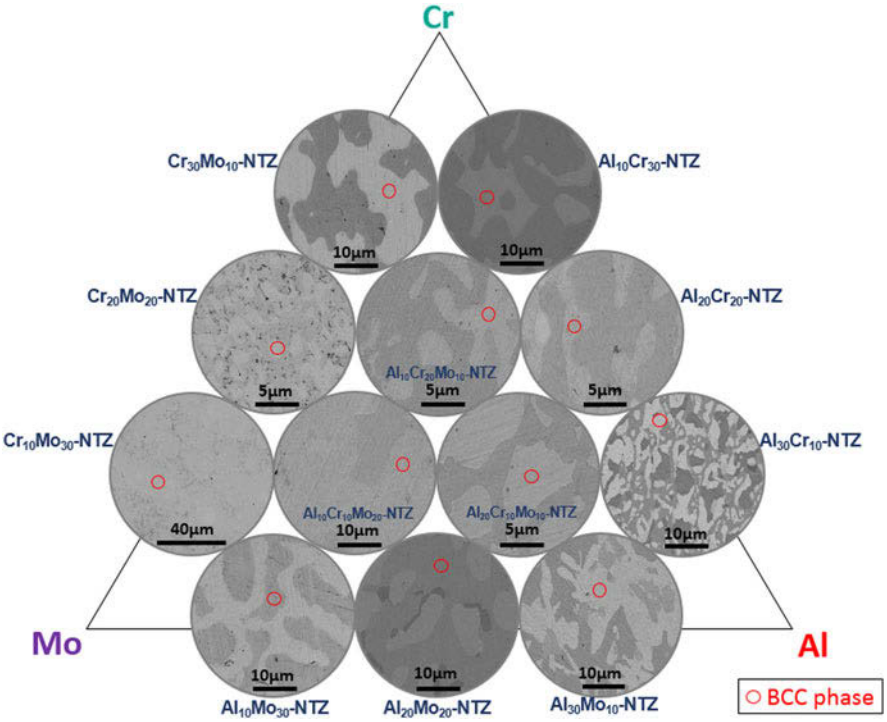


FIG. 2. A combinatorial library of SEM images of the homogenized $Al_xCr_yMo_z$-NTZ samples.

behavior of the $Al_xCr_yMo_z$-NTZ HEA system to understand the variation of the alloying elements.

II. EXPERIMENTAL

The metal sources (granules/sheets) of Al (99.9%), Cr (99.9%), Mo (99.9%), Nb (99.9% pure), Ti (99.9%), and Zr (99.2%) were melted using an arc melting furnace (ACM-01, DAIA-VACUUM, Tokyo, Japan) to form $Al_xCr_yMo_z$-NTZ. Details of the compositions of the samples are given in Table I.

Melting was done in an inert gas (Ar) atmosphere (10^{-4}–10^{-5} mbar). A high-purity Ti-sponge was also used as an oxygen getter to minimize oxidation. Water-cooled copper molds were used as a melting pit to prevent the melting of the mold and to reduce the temperature at a cooling rate up to 100 K/s after the melting process had completed. All the samples were remelted at least five times to enhance the chemical homogeneity. Further homogenization was carried out by heat treating the samples at 1200 °C up to 24 h in an Ar atmosphere, as per previous practices.[25–27] A microstructural analysis of the $Al_xCr_yMo_z$-NTZ samples was carried out by X-ray diffraction (XRD, D/MAX-2500, Rigaku, Tokyo, Japan) and scanning electron microscopy coupled with energy dispersive spectroscopy (SEM-EDS, FEI Magellan 400, Hillsboro, Oregon). The effect of varying the composition on the isothermal oxidation resistance of $Al_xCr_yMo_z$-NTZ (2×4 mm^2 samples) was analyzed via a thermogravimetric analysis (TGA) at 1000 °C in air up to 1 h. An oxidation test at 1000 °C for 20 h was conducted to assess the mass change and the integrity of the $Al_xCr_yMo_z$-NTZ HEA samples after extended oxidation.

III. RESULTS AND ANALYSIS

The SEM analysis of the homogenized $Al_xCr_yMo_z$-NTZ HEAs reveals a coarse granular microstructure[7] with clear boundaries between the solid solution and the secondary phases as shown in Fig. 2. The secondary phases in refractory HEAs have been identified as various kinds of intermetallic phases including the hexagonal C14 Laves phase[2–4,7,19,28,29] and the cubic C15 Laves phase.[30] In the AlCrNbTiV[9] and AlCrMoNbTi[31] HEAs, the Laves phase compositions of (Nb, Ti)(Al, Cr, V)$_2$[9] and (Nb, Ti)(Al, Cr, Mo)$_2$[31] were described as having AB_2 stoichiometry in which A and B are elements with greater and lesser atomic sizes, respectively.[9]

The coupled analyses of the SEM microstructures with XRD and EDS provided useful information about the structures and compositions of the constituent phases as illustrated using the cases of $Cr_{30}Mo_{10}$-NTZ and $Al_{10}Cr_{30}$-NTZ in Fig. 3. XRD patterns of the homogenized $Cr_{30}Mo_{10}$-NTZ and $Al_{10}Cr_{30}$-NTZ samples show major BCC structures along with secondary phases. A Cr_2Zr-type C15 Laves phase was identified in $Cr_{30}Mo_{10}$-NTZ and an Al_3Zr_5-type intermetallic phase was identified in $Al_{30}Cr_{10}$-NTZ as the major secondary

phases, respectively. The chemical compositions of the BCC phases and the intermetallic phases in the $Cr_{30}Mo_{10}$-NTZ and $Al_{10}Cr_{30}$-NTZ samples measured by the EDS analysis were presented in the insets of Fig. 3, respectively. Table II shows the chemical composition of the BCC phases observed in the twelve $Al_xCr_yMo_z$-NTZ samples. In general, the concentrations of Nb and Ti are higher than 20 at.%, while the concentrations of Zr are lower than 20 at.% in the BCC phases in the $Al_xCr_yMo_z$-NTZ samples.

The likelihood of intermetallic phases and solid solution phases in $Al_xCr_yMo_zNbTiZr$ HEAs was estimated by comparing the atomic size difference (δ_r), the Allen electronegativity difference (ΔX_{Allen}),[25] and the average value of the *d*-orbital energy level ($\overline{Md}$) with the intermetallic phase formation criteria. Yurchenko et al. analyzed that Laves phase formation is observed in HEAs having an atomic size difference (δ_r) greater than 5% and an Allen electronegativity difference (ΔX_{Allen}) exceeding 7%.[25]

We calculated δ_r and ΔX_{Allen} for $Al_xCr_yMo_zNbTiZr$ and these values are given in Table III. For most $Al_xCr_yMo_zNbTiZr$ HEAs, the δ_r values are greater than 5%, except for Al_xMo_z-NTZ HEAs. The ΔX_{Allen} values of $Al_xCr_yMo_zNbTiZr$ exceed 7% for most compositions. An increase in the Cr content leads to an increase in ΔX_{Allen}. However, for $Al_{20}Mo_{20}$-NTZ, $Al_{10}Mo_{30}$-NTZ, and $Cr_{10}Mo_{30}$-NTZ, $\Delta X_{Allen} \leq 7\%$. The values of δ_r and ΔX_{Allen} suggest intermetallic phases in most $Al_xCr_yMo_zNbTiZr$ HEAs. While $Al_{20}Mo_{20}$-NTZ and $Al_{10}Mo_{30}$-NTZ have $\delta_r < 5$ and $\Delta X_{Allen} \leq 7\%$, a microstructural analysis exhibits the existence of intermetallic phases similar to the other $Al_xCr_yMo_zNbTiZr$ HEAs as shown in Fig. 2.

Morinaga et al. analyzed that superalloys having an average value of the energy level of the *d*-orbital ($\overline{Md}$) higher than 0.915 eV exhibit the σ phase.[32] Lu et al.[33] applied Morinaga's approach to HEAs and observed that

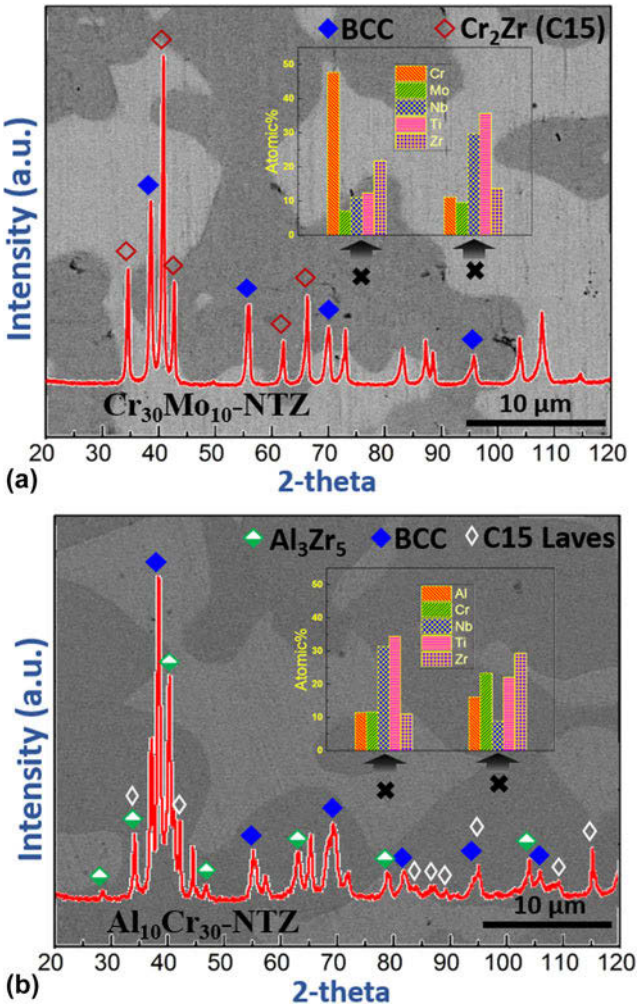


FIG. 3. Microstructural analyses of (a) $Cr_{30}Mo_{10}$-NTZ and (b) $Al_{10}Cr_{30}$-NTZ by SEM, XRD, and EDS.

TABLE II. EDS analysis of the BCC phases in the $Al_xCr_yMo_z$-NTZ samples.

Sample name	Composition (at.%)					
	Al	Cr	Mo	Nb	Ti	Zr
$Al_{20}Cr_{10}Mo_{10}$-NTZ	19.4	6.0	7.8	29.0	30.0	7.8
$Al_{10}Cr_{20}Mo_{10}$-NTZ	8.2	10.3	9.5	28.3	33.7	10.0
$Al_{10}Cr_{10}Mo_{20}$-NTZ	8.0	6.0	14.7	27.1	32.0	12.0
$Al_{30}Mo_{10}$-NTZ	35.1	...	6.8	24.5	17.8	15.9
$Al_{20}Mo_{20}$-NTZ	20.2	...	15.1	21.4	27.2	16.1
$Al_{10}Mo_{30}$-NTZ	6.0	...	31.9	22.9	30.2	9.0
$Al_{30}Cr_{10}$-NTZ	34.6	3.0	...	26.9	21.7	13.8
$Al_{20}Cr_{20}$-NTZ	19.5	8.6	...	27.7	34.1	10.2
$Al_{10}Cr_{30}$-NTZ	11.4	11.6	...	31.4	34.5	11.1
$Cr_{30}Mo_{10}$-NTZ	...	11.2	9.6	29.8	35.7	13.7
$Cr_{20}Mo_{20}$-NTZ	...	9.0	17.4	26.9	33.7	13.0
$Cr_{10}Mo_{30}$-NTZ	...	5.2	27.1	25.2	19.8	12.5

TABLE III. Atomic size mismatch (δ_r), Allen electronegativity difference (ΔX_{Allen}), and average values of *d*-orbital energy level ($\overline{Md}$) calculated for $Al_xCr_yMo_zNbTiZr$.

Sample name	δ_r, %	ΔX_{Allen}, %	$\overline{Md}$, eV
$Al_{20}Cr_{10}Mo_{10}$-NTZ	5.93	8.01	2.85
$Al_{10}Cr_{20}Mo_{10}$-NTZ	7.12	8.35	2.77
$Al_{10}Cr_{10}Mo_{20}$-NTZ	6.32	7.39	2.81
$Al_{30}Mo_{10}$-NTZ	4.31	7.65	2.92
$Al_{20}Mo_{20}$-NTZ	4.60	7.00	2.89
$Al_{10}Mo_{30}$-NTZ	4.84	6.30	2.85
$Al_{30}Cr_{10}$-NTZ	5.75	8.57	2.88
$Al_{20}Cr_{20}$-NTZ	6.99	8.91	2.81
$Al_{10}Cr_{30}$-NTZ	7.94	9.23	2.73
$Cr_{30}Mo_{10}$-NTZ	8.02	8.69	2.70
$Cr_{20}Mo_{20}$-NTZ	7.22	7.77	2.74
$Cr_{10}Mo_{30}$-NTZ	6.26	6.73	2.78

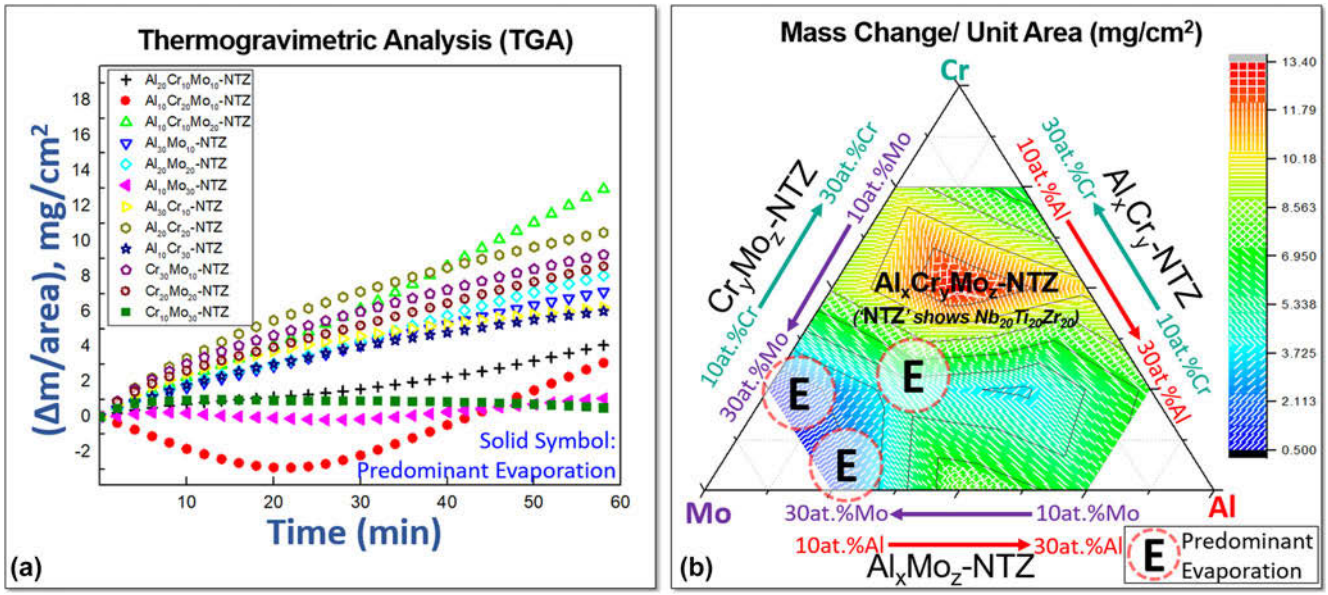


FIG. 4. (a) TGA curves and (b) total mass change/unit area after 1 h of oxidation at 1000 °C in a pseudo-ternary map of homogenized $Al_xCr_yMo_z$-NTZ.

the HEAs with $\overline{Md} > 1.09$ exhibit stable intermetallic phases with few exceptions. As per the methods given in the literature,[32,33] we calculated the average value of the energy level of the *d*-orbital $(\overline{Md})$ for $Al_xCr_yMo_zNbTiZr$, as shown in Table III. According to the $\overline{Md}$ values, secondary phases are expected to appear in all compositions of $Al_xCr_yMo_zNbTiZr$.

It has been reported that the oxidation of HEA yields a mixture of various oxides due to multiple principal elements.[13,21,34] The literature suggests the presence of various oxides, including TiO_2, $Nb_2Zr_6O_{17}$, $TiNb_2O_7$, ZrO_2, $NbCrO_4$, Cr_2O_3, NbO,[20] Cr_2O_3,[35] Al_2O_3,[36] $Cr_2Ti_5O_{13}$, $Nb_3Cr_2O_{10}$, $Nb_2Zr_8O_{21}$, Ti_2ZrO_6, $Ti_{0.8}O_2$, Cr_2TiO_5, and Nb_2O_5[21] in refractory HEAs. To analyze the effect of various constituent elements and quantify the oxidation resistance, the TGA method can be used for HEAs.[20,21,35,37] The change in the mass of the $Al_xCr_yMo_z$-NTZ HEA samples during oxidation and the change in the oxidation rate were estimated.[38] The samples were exposed to air at 1000 °C and the change in the mass was measured, as presented in Fig. 4(a). The mass change per unit area after one hour of oxidation at 1000 °C is mapped on a pseudo-ternary diagram as shown in Fig. 4(b).

The HEA samples with a higher concentration of Mo showed a negative slope of the TGA curves [solid symbols in Fig. 4(a)]. For example, the masses of $Al_{10}Mo_{30}$-NTZ, $Cr_{10}Mo_{30}$-NTZ, and $Al_{10}Cr_{10}Mo_{20}$-NTZ decreased at a certain period during the oxidation experiment. This behavior can be attributed to the formation of a volatile oxide, as has been reported for Mo-rich refractory alloys.[39] The evaporation during the oxidation of $Al_{10}Mo_{30}$-NTZ, $Cr_{10}Mo_{30}$-NTZ, and $Al_{10}Cr_{10}Mo_{20}$-NTZ may also be related to the significant porosity in the oxide layers.[40] The microstructural analysis of the oxidized samples revealed porous oxide layers for volatile oxidation in $Al_xCr_yMo_z$-NTZ. The representative microstructures of the porous ($Cr_{10}Mo_{30}$-NTZ) and dense oxide layers ($Al_{30}Cr_{10}$-NTZ, respectively) are shown in Figs. 5(a) and 5(b), respectively. Mixed phases were observed in the oxide layers $Al_{30}Cr_{10}$-NTZ, but no Al-rich layers were found in the oxide layers.

Figure 6 shows the variation in the surface oxide thicknesses after the TGA. The oxide layer thickness in the $Al_xCr_yMo_z$-NTZ HEA system varies between 60 and 600 μm. Although a direct comparison of the oxide thicknesses of the HEA compositions is less meaningful due to volatile oxidation in some samples, it is important to note the effect of Al on reducing the oxidation thickness in Mo-free Al_xCr_y-NbTiZr as compared to that in Mo-bearing HEAs. $Al_{10}Cr_{30}$-NTZ shows a 107-μm-thick oxide scale that remains unchanged upon increasing the Al content to 20 at.%. A further addition of Al to yield $Al_{30}Cr_{10}$-NTZ shows a significant reduction in the oxide thickness to 79 μm, as shown in Fig. 6.

The formation of stable oxide films reduces the contact between oxygen and the bare metal surface and controls the oxidation rate. To determine the protectiveness of the oxide scale on $Al_xCr_yMo_z$-NTZ HEA, the mass change versus time was plotted on a log–log scale graph [Fig. 7 (a)] and the slope was determined [Fig. 7(b)]. The protective oxide layer is observed when the log–log graph of the mass change versus time shows a slope (empirical exponent, *n*) of 0.5.[21] The plot of mass gain

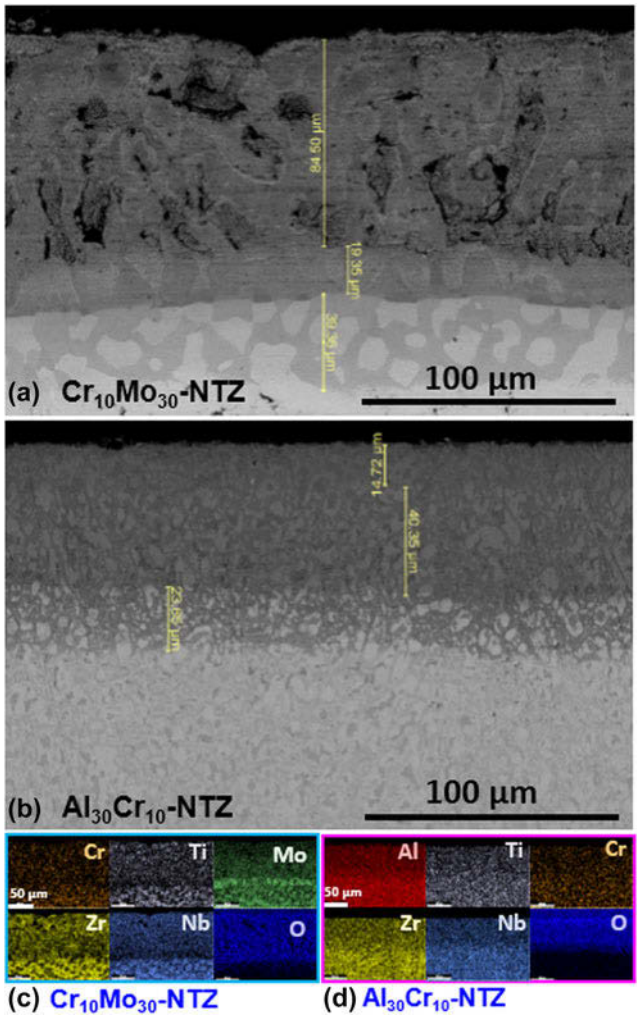


FIG. 5. Microstructures of (a) $Cr_{10}Mo_{30}$-NTZ and (b) $Al_{30}Cr_{10}$-NTZ, and EDS analysis results of (c) $Cr_{10}Mo_{30}$-NTZ and (d) $Al_{30}Cr_{10}$-NTZ of the oxide layers of $Al_xCr_yMo_z$-NTZ.

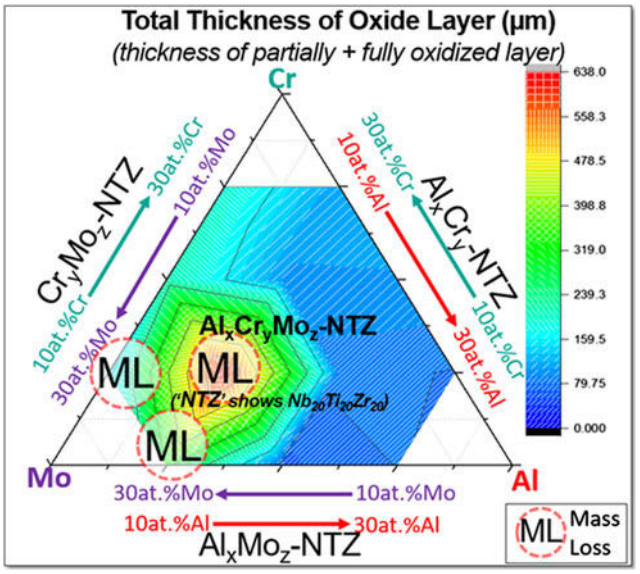


FIG. 6. Variation in the thickness of the oxide layer of $Al_xCr_yMo_z$-NTZ.

per unit surface area of $CrMo_{0.5}NbTa_{0.5}TiZr$ on a logarithmic scale shows two distinct features. First, the oxidation started at a higher rate and exhibited $n = 0.85$ (up to 25 min), after which the development of the oxide scale reduced the rate of the reaction and n became 0.5 (which remained at that value for up to 60 min).[21] The oxidation of NbTiVZr at 1000 °C proceeded with $n = 0.9$ until the HEA sample was fully oxidized within 8 h.[20] By contrast, the oxidation of CrNbTiZr was controlled by an oxygen-alloy interfacial reaction in the initial stage up to 1.8 h, with n remaining at 0.74, after which the diffusion of oxygen through the oxide scale started, controlling the rate of the oxidation reaction, with n reduced to 0.34 and remaining there until 100 h of oxidation.[20] To compare the oxidation behaviors of these reported HEAs with that of $Al_xCr_yMo_z$-NTZ, the slopes of the first 10 min and the last 10 min were compared, as shown in Fig. 7(b), while the values for the samples which presented negative slopes were excluded. During the first ten minutes of oxidation, the $Al_xCr_yMo_z$-NTZ HEA shows n values ranging from 0.81 to 0.91.

During the last ten minutes of oxidation, the n value for $Al_xCr_yMo_z$-NTZ changed from the values during the initial 10 min. The slopes of the Mo-containing HEAs did not decrease much, as shown in Fig. 7(b). Some of them showed increased slopes, as shown in $Al_{20}Cr_{10}Mo_{10}$-NTZ and $Al_{10}Cr_{20}Mo_{10}$-NTZ. The Mo-free HEA samples of $Al_{30}Cr_{10}$-NTZ, $Al_{20}Cr_{20}$-NTZ showed n values close to 0.5, suggesting the formation of protective oxide layers. The slope of ~0.5 for these alloys also suggests that the degree of protectiveness of the oxide layer on these alloys is comparable to that of $CrMo_{0.5}NbTa_{0.5}TiZr$[21] and is better than those of NbTiVZr and CrNbTiZr.[20] HEAs having higher Al contents (within their respective groups) showed lower n values during the last ten minutes of oxidation as well. Hence, Al is observed to be effective in forming a passive oxide layer.

Mixtures of oxides are identified by an analysis of the XRD patterns of oxidized HEA samples (Fig. 8). For example, the oxide scale on $Al_{20}Mo_{20}$-NTZ shows $Nb_2Zr_6O_{17}$, $AlNbO_4$, and TiO_2. $Nb_2Zr_6O_{17}$ and $ZrTiO_4$ are observed in the oxide scale of $Cr_{20}Mo_{20}$-NTZ and $Al_{30}Cr_{10}$-NTZ, respectively. In addition to these, other complex oxides may also be present. The compositions of the oxide scale are similar to those in other studies in which the formation of similar oxides after the oxidation of refractory HEAs was reported.[20,21,35,36]

To analyze the response of these HEAs to prolonged oxidation, the samples were subjected to oxidation in static air (at 1000 °C) for 20 h. After 20 h of oxidation, the $Al_{30}Cr_{10}$-NTZ, $Al_{20}Cr_{10}Mo_{10}$, and $Al_{10}Cr_{30}$-NTZ

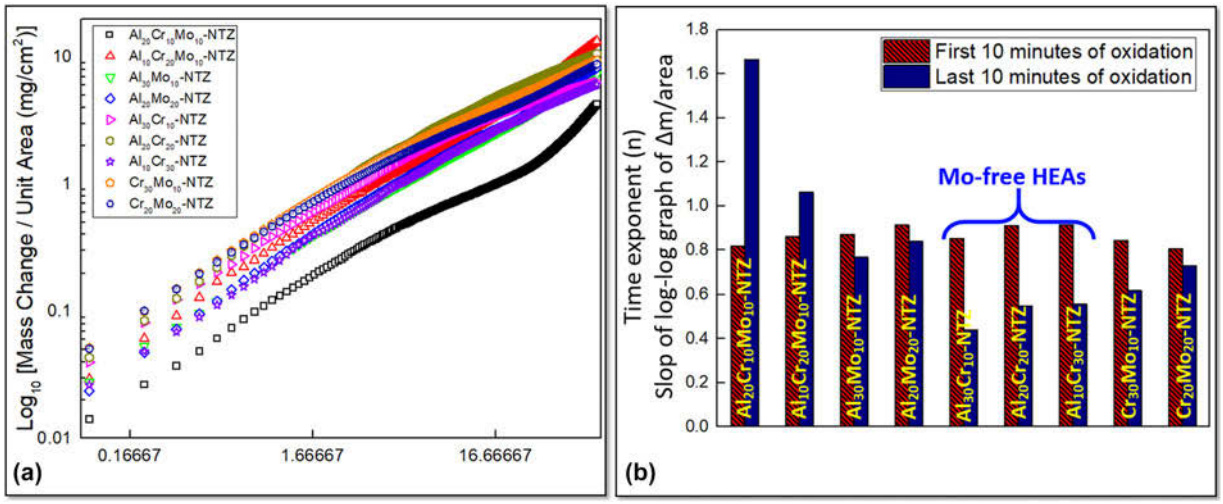


FIG. 7. (a) TGA curves on a log–log scale and (b) time exponent [slopes of $\log_{10}(\Delta m/\text{area})$ versus $\log_{10}(\text{time})$ curve] during the first and last ten minutes of oxidation.

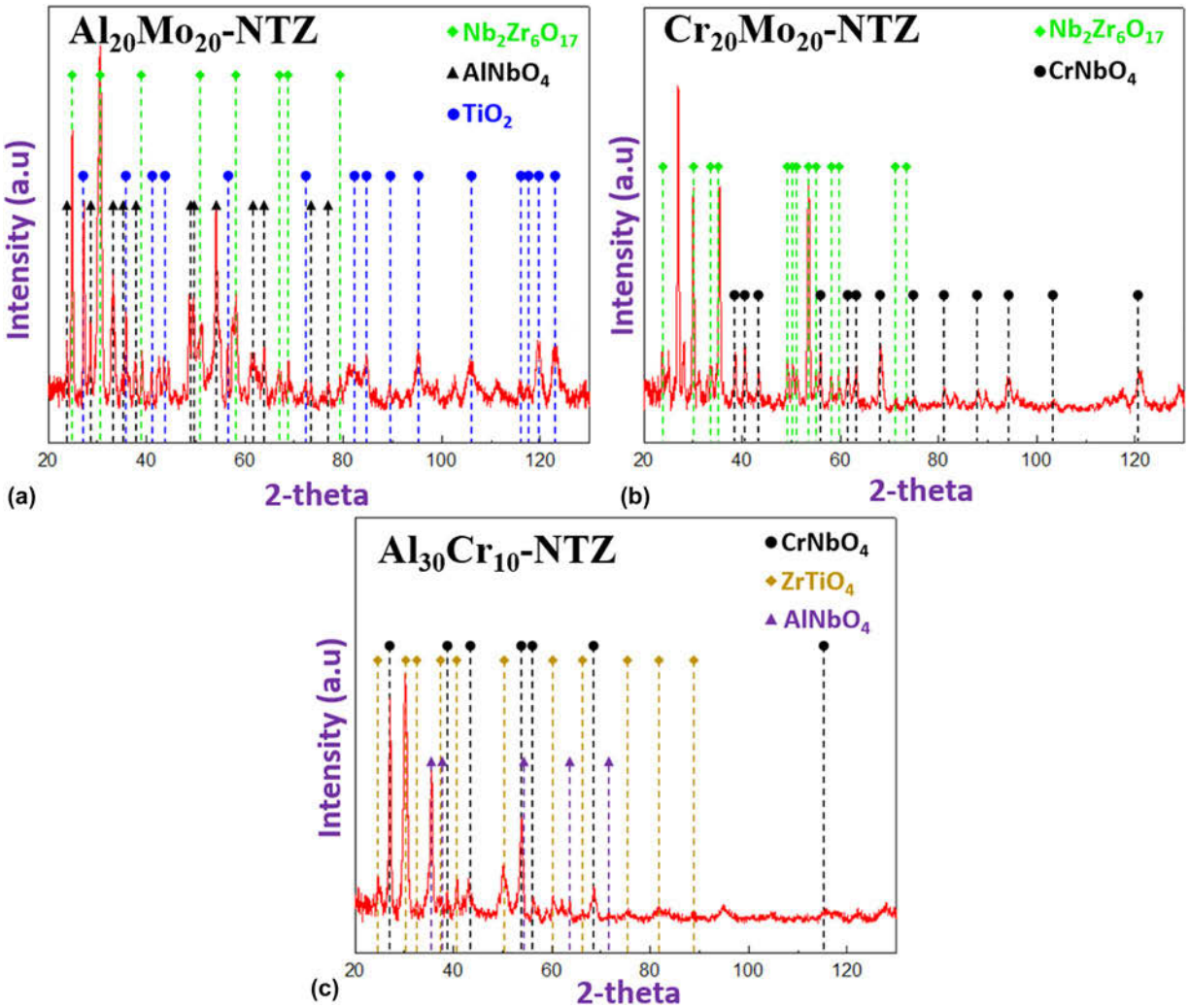


FIG. 8. XRD analysis results of the oxide layers on three $Al_xCr_yMo_z$-NTZ HEAs [(a) $Al_{20}Mo_{20}$-NTZ, (b) $Cr_{20}Mo_{20}$-NTZ, and (c) $Al_{30}Cr_{10}$-NTZ] oxidized at 1000 °C for 1 h.

TABLE IV. Comparison of the mass gain after prolonged oxidation of $Al_{20}Cr_{10}Mo_{10}$-NTZ, $Al_{30}Cr_{10}$-NTZ, and $Al_{10}Cr_{30}$-NTZ with relevant HEAs and refractory alloys.

Sample name	Mass gain after oxidation, (1000 °C, 20 h), mg/cm^2
$Al_{20}Cr_{10}Mo_{10}$-NTZ	21
$Al_{30}Cr_{10}$-NTZ	20
$Al_{10}Cr_{30}$-NTZ	39
AlCrMoNbTi[31,36]	3
$AlMo_{0.5}NbTa_{0.5}TiZr$[21]	40
9Al–Nb–8Si–10Ti[47]	180
9Al–Nb–10Si–10Ti[47]	90
11Al–Nb–6Si–15Ti[47]	50
11Al–Nb–8Si–15Ti[47]	50

samples remained intact and their oxide layers remained adhered to the surface. The other samples were either fully oxidized or their oxide layers were detached. The comparison of oxidation resistance capabilities of $Al_{30}Cr_{10}$-NTZ, $Al_{20}Cr_{10}Mo_{10}$, and $Al_{10}Cr_{30}$-NTZ demonstrates higher oxidation resistance in these samples relative to those of several Nb-based alloys (Table IV). Furthermore, we could find only two reports of HEAs with significant oxidation resistance: $AlMo_{0.5}NbTa_{0.5}TiZr$[21] and AlCrMoNbTi.[36] The mass gain in AlCrMoNbTi[36] was quite low, suggesting quite stellar oxidation resistance. However, $Al_{30}Cr_{10}$-NTZ, $Al_{20}Cr_{10}Mo_{10}$, and $Al_{10}Cr_{30}$-NTZ reveal better oxidation resistance capabilities than $AlMo_{0.5}NbTa_{0.5}TiZr$. To quantify the volatile oxidation and evaporation of Mo-containing HEAs, TGA should be coupled with mass spectrometry.[41]

Recent publications pertaining to refractory HEAs report samples with higher mechanical strength and strain levels, but all of those alloys possess W,[11,42,43] Ta,[11,42,44] V,[15,45] and/or Hf,[12,42,44,46] with deleteriously high-density levels (due to Ta and W), high costs (due to Hf and Ta), or oxidation (due to V). Therefore, the commercial feasibility of such HEAs is uncertain. $Al_{20}Cr_{10}Mo_{10}$-NTZ and $Al_{30}Cr_{10}$-NTZ showed good oxidation resistance levels without such deleterious elements. This forecasts the potential high-temperature applications of these HEAs; therefore, their mechanical properties need to be analyzed in detail in the near future.

IV. CONCLUSION

Here, arc-melted and homogenized $Al_xCr_yMo_z$-NTZ HEAs showed a BCC structure along with intermetallic phases including Al_3Zr_5, cubic C15 and hexagonal C14 Laves phases. Oxidation at 1000 °C for 1 h led to a 0.5–13.38 mg/cm^2 gain in the mass. The Mo-rich HEAs were associated with mass losses due to volatile oxidation. Increasing the concentration of Al resulted in a reduction in the rate of the oxidation reaction, thus demonstrating the role of Al in improving the oxidation resistance of $Al_xCr_yMo_z$-NTZ. The time exponent of Mo-free HEAs was found to be closer to 0.5, ensuring the protectiveness of the oxide layer. $Al_{30}Cr_{10}$-NTZ showed a weight gain of 20 mg/cm^2 after prolonged oxidation up to 20 h in air at 1000 °C. The enhanced oxidation resistance capabilities of $Al_{30}Cr_{10}$-NTZ forecast its potential for use in high-temperature applications.

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